ic_autolab round 16 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T06:42:40Z at git `1d09988e20cc91cf05990398bed753359216ff58`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-1c867a` (`python3 tools/pcc/pcc.py standing PCC-20261104-1c867a`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20	**improvement**	wsum-b5-n3	0.605 [0.575, 0.637]	0.584 [0.553, 0.618]	0.478 [0.441, 0.519]	sel < inc	5.384 / 5.609 / 7.638	8.893 / 9.601 / 15.971	1.487 / 1.592 / 2.538	3.6x / 3.5x / 3.0x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

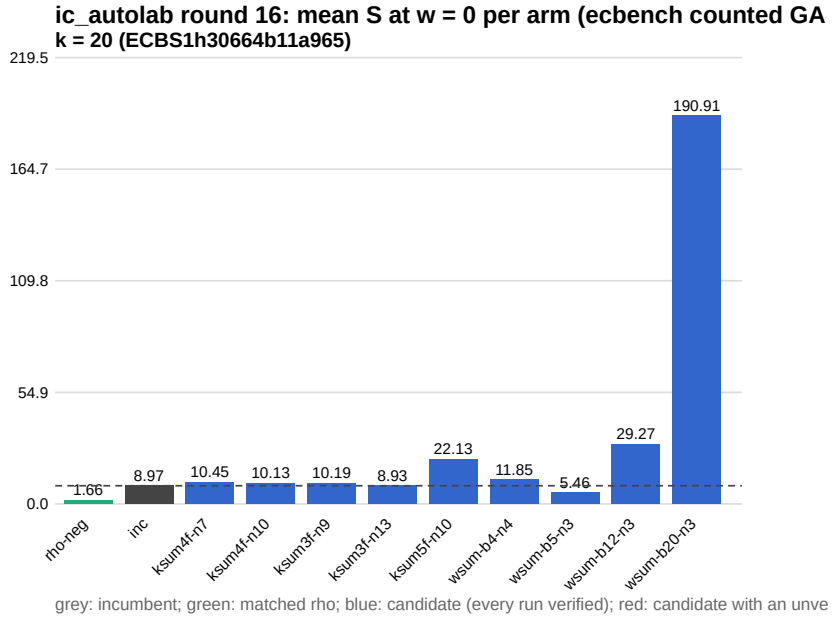
- k = 20: compare ops.status = ok, admissible = true; sel / inc = 0.6054 [0.575, 0.637] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 20: session `ECBS1h30664b11a965` (run exit 0, verify exit 0, receipt sha256 `65e232a89888a4ba7f54d598b296572231c5df8e1a47de6b6a67f19e85b6a2b0`)

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	32	32	1.660 / 1.783 / 2.886	matched reference
`inc`	ksum3	11	`prime-projected-abscissa:size=11,scan_cap=176`	`ksum:m=3`	32	32	8.970 / 9.710 / 16.368	incumbent
`ksum4f-n7`	ksum4f	7	`prime-projected-abscissa:size=7,scan_cap=112`	`ksum:m=4,free=1`	32	32	10.452 / 11.535 / 21.288	alternative ksum4f (best S(0) 8.282 at N = 8): N = 0.85 x 8
`ksum4f-n10`	ksum4f	10	`prime-projected-abscissa:size=10,scan_cap=160`	`ksum:m=4,free=1`	32	32	10.129 / 10.433 / 13.173	alternative ksum4f (best S(0) 8.282 at N = 8): N = 1.2 x 8
`ksum3f-n9`	ksum3f	9	`prime-projected-abscissa:size=9,scan_cap=144`	`ksum:m=3,free=1`	32	32	10.193 / 11.240 / 20.671	alternative ksum3f (best S(0) 9.043 at N = 11): N = 0.85 x 11
`ksum3f-n13`	ksum3f	13	`prime-projected-abscissa:size=13,scan_cap=208`	`ksum:m=3,free=1`	32	32	8.927 / 9.457 / 14.222	alternative ksum3f (best S(0)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							9.043 at N = 11): N = 1.2 x 11
ksum5f-n10	ksum5f	10 prime-projected-abscissa:size=10,scan_cap=160	ksum:m=5,free=1	32	32	22.132 / 22.738 / 28.198	alternative ksum5f (best S(0) 9.379 at N = 8): N = 1.2 x 8
wsum-b4-n4	wsum-b4	4 prime-projected-abscissa:size=4,scan_cap=64	wsum:coef=4,free=1	32	32	11.851 / 11.941 / 12.745	exploration: family wsum-b4 untried at k = 20, N = 0.0001 x N_a
wsum-b5-n3	wsum-b5	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=5,free=1	32	32	5.459 / 5.706 / 7.925	exploration: family wsum-b5 untried at k = 20, N = 0.0001 x N_a
wsum-b12-n3	wsum-b12	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=12,free=1	32	32	29.267 / 29.840 / 34.995	exploration: family wsum-b12 untried at k = 20, N = 0.0001 x N_a
wsum-b20-n3	wsum-b20	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=20,free=1	32	32	190.912 / 200.838 / 290.173	exploration: family wsum-b20 untried at k = 20, N = 0.0001 x N_a



train screen

Selection

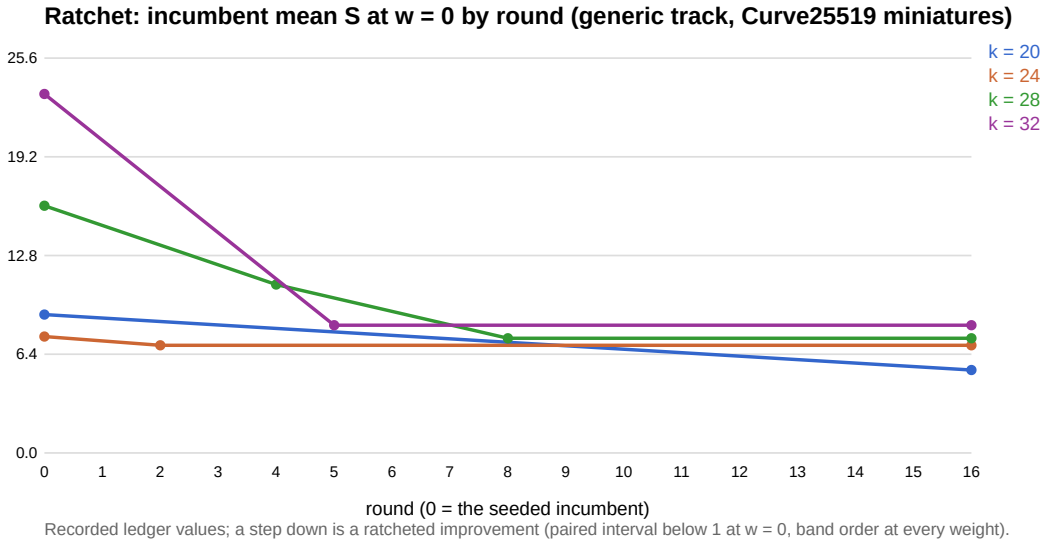
- k = 20: challenger `wsum-b5-n3` (Pareto alternative at w = 1: `wsum-b5-n3`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
20	`ECBS1h458afdb84a20`	0	0	`3f6ccf922ffe57b4946303075ac33e0e144e60e1fa096858a243be80fadb648c`	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum-b5-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=5,free=1`)	16	5.384 / 5.609 / 7.638	3.6x / 3.5x / 3.0x
24	`ksum4-n12` (`prime-projected-abscissa:size=12,scan_cap=192` + `ksum:m=4`)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	`ksum4-n18` (`prime-projected-abscissa:size=18,scan_cap=288` + `ksum:m=4`)	8	7.445 / 7.910 / 12.100	7.8x / 7.6x / 6.5x
32	`ksum4-n26` (`prime-projected-abscissa:size=26,scan_cap=416` + `ksum:m=4`)	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x



ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.